Change Knowledge-Guided Vision-Language Remote Sensing Change Detection

Jiahao Wang, Fang Liu, *Senior Member, IEEE*, Licheng Jiao, *Fellow, IEEE*, Hao Wang, Shuo Li, *Member, IEEE*, Lingling Li, *Senior Member, IEEE*, Puhua Chen, *Senior Member, IEEE*, Xu Liu, *Senior Member, IEEE*, and Wenping Ma, *Senior Member, IEEE*

***Abstract*— Remote sensing image change detection plays a critical role in applications like video surveillance and geographic information systems. However, existing binary and semantic change detection (SCD) methods often rely solely on visual information, neglecting language information, which limits interpretability and the ability to provide specific change details. This work proposes the change knowledge-guided vision-language (VL) remote sensing change detection (CKCD) method to address these limitations. By introducing change knowledge as language information, CKCD enhances semantic understanding and change detail representation. A cross-modal affinity (CMA) module is designed to effectively fuse visual and textual features, improving information complementarity and fusion coherence. CKCD further enhances data utilization efficiency by merging change area detection and change category information into a single output through end-to-end learning. This design reduces redundant data representations and simplifies the detection process, leading to a more compact and efficient use of the input data without requiring additional branches or multiple output heads. Experimental results demonstrate consistent performance improvements over traditional methods across multiple change detection datasets.**

***Index Terms*— Change category information, change detection, change knowledge, visual and textual modalities.**

I. Introduction

REMOTE sensing image change detection, as a crucial application of remote sensing technology, focuses on accurately capturing and quantifying the spatiotemporal changes in surface features. It holds immense value in urban planning, environmental monitoring, efficient resource management, and rapid disaster response [1], [2], [3], [4], [5], [6].

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Joint International Research Laboratory of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: f63liu@163.com).

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Currently, the core of change detection tasks revolves around two primary branches: binary change detection (BCD) and semantic change detection (SCD). However, both approaches predominantly rely on single visual modality information, overlooking the potential significance of language information. This reliance leads to a lack of multidimensional information, resulting in limited interpretability and transparency when dealing with complex scenarios. Furthermore, it restricts the adaptability and transferability of these methods, causing them to perform poorly when faced with new application scenarios or datasets. In addition, these methods often struggle to capture global information and inherent logic, rendering them inadequate to handle the complex and dynamic natural phenomena in remote sensing images.

BCD methods, represented by FC-EF [7] and its subsequent studies [8], [9], [10], [11], [12], [13], [14], [15], primarily focus on generating binary change maps, as illustrated in Fig. 1(a). Their simplicity and efficiency have facilitated widespread application in remote sensing. However, these methods only detect the presence or absence of change, without providing detailed semantic descriptions of the change types, which is particularly limiting in applications requiring fine-grained classification. To address these limitations, SCD methods, such as HRSCD [16] and Bi-SRNet [17], have been proposed to incorporate semantic category information alongside BCD. These approaches employ multioutput heads to achieve simultaneous recognition of change regions and categories, as shown in Fig. 1(b). However, the multibranch design increases model complexity and may lead to suboptimal results when facing ambiguous semantic categories. Recently, vision-language (VL) models have shown promising results in remote sensing tasks by leveraging textual descriptions for enhanced semantic understanding. Models such as CLIP [18] and BLIP [19] have demonstrated the potential of language information in improving classification and detection performance. However, existing CD approaches have yet to fully explore the benefits of VL models for SCD, particularly in complex environments where both visual and language information can provide complementary insights. The absence of language information integration can limit the model's ability to understand the semantic meaning behind changes and the relationships between categories, which, along with other factors, may lead to inconsistencies in detection and recognition in complex environments, thereby restricting the model's transferability and generalization capabilities [20], [21], [22].

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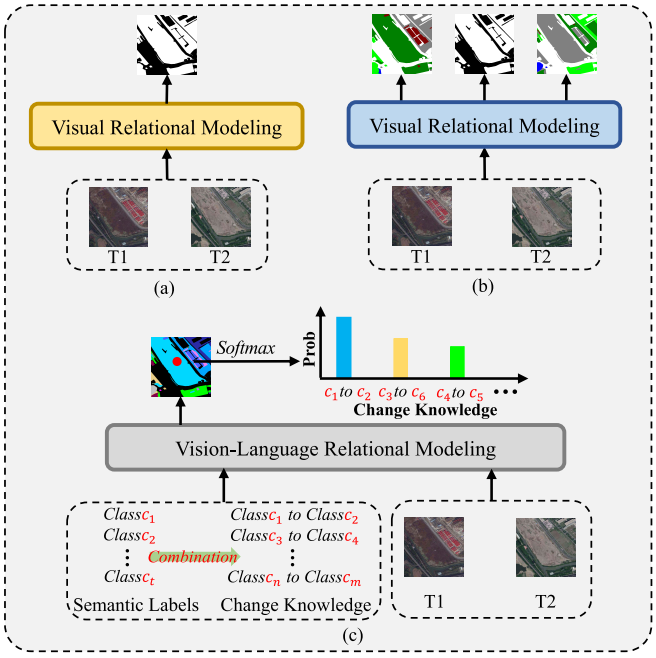


Fig. 1. Illustrations of three different frameworks for the change detection task. (a) BCD that generates a binary change map. (b) SCD with three decoders. (c) Proposed CKCD method.

To overcome these limitations, we propose the change knowledge-guided VL remote sensing change detection (CKCD) method. CKCD introduces change knowledge, extracted from existing SCD datasets, as semantic labels to enrich the model's understanding of change types. This language information is directly integrated into the detection process, improving semantic comprehension and decision transparency. By incorporating this change in knowledge, our model significantly enhances decision-making transparency, making the detection results more intuitive and easier to understand and interpret. In addition, we designed an integrated approach that consolidates the previously fragmented multioutput heads of SCD into a single output, achieving a truly end-to-end detection process. This improvement allows the output head to simultaneously provide both the precise location of change regions and the corresponding change category information, as illustrated in Fig. 1(c), greatly enriching the dimensions of information transmission. This method offers several advantages.

1) *Enhanced Richness of Information Expression:* Compared to traditional BCD, our method outputs specific change category information, providing a more detailed description of change characteristics and a higher level of semantic understanding, thereby significantly increasing the expressiveness of the information.
2) *High Transferability and Flexibility:* The new method demonstrates strong adaptability thanks to its modular design. While traditional binary detection methods require complex structural adjustments to adapt to new tasks, our method can easily transfer to new scenarios with a simple adjustment of the output categories, offering unprecedented flexibility and convenience for practical applications.
3) *Optimized End-to-End Learning:* By seamlessly combining change region detection with change category recognition into a single output head, the model better understands the intrinsic relationship between changes and categories, leading to more accurate and efficient change detection.

Specifically, in this work, we have designed the CKCD method, which effectively leverages change knowledge to better align the proposed approach with the requirements of change detection tasks. Building on this foundation, we introduce the cross-modal affinity (CMA) module to bridge the gap between visual and textual modalities. This not only enhances the complementarity of information but also ensures the coherence of the fusion process, thereby improving the accuracy and robustness of the detection. The contributions of the work are summarized as follows.

1) We propose the CKCD method, which integrates change knowledge as semantic labels to improve the semantic clarity and detection consistency in remote sensing change detection tasks.
2) We introduce the CMA module to address the discrepancies between visual and textual modalities, thereby enhancing the complementarity of information and ensuring consistency in the fusion process. By integrating change knowledge and utilizing the CMA module, the proposed CKCD method significantly improves the accuracy and robustness of remote sensing change detection, outperforming existing methods.
3) We validate the effectiveness of the CKCD method through extensive experiments on five datasets: the restructured semantic category datasets SECOND and Hi-UCD for SCD tasks, and the BCD datasets LEVIR, WHU, and CDD. The results consistently demonstrate superior accuracy, robustness, and generalization capabilities compared to state-of-the-art methods.

The remainder of this article is organized as follows. Section II introduces related work. Section III provides a detailed description of the proposed CKCD method, including the architecture and the components such as the CMA module and the fusion strategies. Section IV outlines the experimental setup, datasets, evaluation metrics used for performance assessment, experimental results, and ablation studies, highlighting the contributions of each component. Finally, Section V concludes the article, and discusses the findings, limitations, and potential future directions.

II. Related Work

A. Binary Change Detection

Deep learning-based BCD methods are typically classified into two main categories: 1) full convolutional network-based change detection [23], [23], [24], [25], [26], [27], [28], [29], [30] and 2) Transformer-based change detection [31], [32], [33], [34], [35], [36]. Many recent BCD methods leverage Transformer structures, achieving remarkable performance on various change detection benchmarks. For instance, the study

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by [31] demonstrated the effectiveness and efficiency of change detection in high-resolution remote sensing images using the ConvNet + Transformer architecture. This structure effectively identifies relevant changes while excluding irrelevant ones by modeling contextual relationships and semantic markers. Unlike the dual-feature extraction structure of BIT [31], [37] operates as a pure Transformer network, employing a concatenated U-shape structure to address the challenge of remote sensing image change detection. SwinSUNet comprises an encoder, a fuser, and a decoder. The encoder is built on the concatenated structure of the hierarchical Swin Transformer [38], facilitating parallel processing of bi-phasic images and the extraction of multiscale features. As VL Model [19], [39], [40], [41], [42], [43], [44], [45], [46] become prevalent, change detection methods based on VL model for remote sensing [47], [48] are emerging. For instance, Wang et al. [49] utilizes remote sensing large models as feature extractors, enhancing change detection performance and building upon the foundation laid by BIT.

B. Semantic Change Detection

The outcome of BCD is a binary change map, providing information solely on whether each pixel point has changed. This information is often insufficiently informative for practical applications. In contrast, SCD enhances the level of detail by incorporating two semantic segmentation decoders atop BCD. This addition provides more comprehensive semantic category information, significantly boosting the method's effectiveness in practical applications. Recently, researchers have developed a series of deep learning-based SCD methods [50], [51], [52]. Daudt et al. [16] employed fully convolutional neural networks to facilitate SCD, enabling the simultaneous execution of change detection and land cover mapping tasks. In addition, the model integrates predicted land cover information to improve the precision of change prediction. On this basis, Yang et al. [53] introduced the asymmetric Siamese networks (ASNs), a three-branch SCD method designed for localizing and identifying semantic changes in multitemporal remote sensing images. One branch is dedicated to detecting change regions, while the remaining branches are employed for semantic map segmentation to identify specific categories of changes. The study presented in [17], the authors propose a bi-temporal semantic reasoning network (Bi-SRNet). Bi-SRNet comprises two types of semantic reasoning blocks, facilitating the reasoning of semantic correlations within a single temporal phase and across temporal phases. The SCanNet method proposed in [54] combines CNN and Transformer to model spatiotemporal dependencies through SCanFormer, introducing a semantic learning scheme to enhance the accuracy of SCD.

Despite existing SCD methods providing semantic category information for change regions, our proposed CKCD method significantly differs in how it expresses and presents this information. CKCD compresses the information typically conveyed through three branches in SCD into a single branch, making the output change category information more intuitive than existing methods' single-category output.

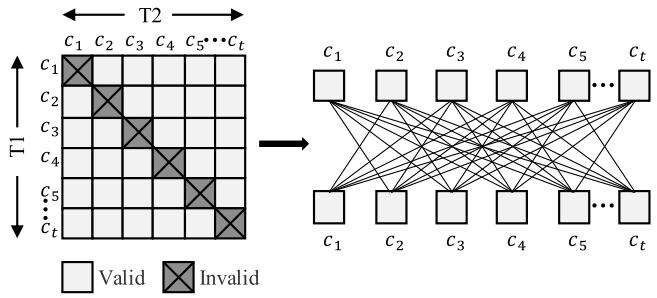


Fig. 2. Illustration of constructing change knowledge. We designate combinations of the same category as "invalid." The tth category is compatible with all other categories, as depicted in the figure on the right.

III. Methodology

This section presents the detailed methodology behind our proposed CKCD framework. First, we outline the problem setting, clearly defining the objectives and challenges of the task. Next, we delve into the overall architecture of CKCD, highlighting the integration of change knowledge to enhance the coherence and effectiveness of remote sensing change detection. Following this, we describe the CMA module, which addresses the alignment between visual and textual modalities, ensuring a complementary and consistent fusion process. Finally, we discuss the training and inference details, providing insights into implementing and optimizing the proposed method.

A. Preliminary

Given a set of dual temporal remote sensing images $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$, the primary goal of traditional BCD is to train a change detection model $\boldsymbol{F}_{\mathrm{BCD}}: \{\boldsymbol{T}_1, \boldsymbol{T}_2\} \rightarrow \boldsymbol{P}_{\mathrm{BCD}}$. This model is designed to predict the binary change map $\boldsymbol{P}_{\mathrm{BCD}}$. In contrast, SCD introduces additional semantic information in the form of category labels. The model's output is extended to $\{\boldsymbol{S}_1, \boldsymbol{S}_2, \boldsymbol{P}_{\mathrm{BCD}}\}$, where $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ represent the semantic maps of the two temporal images. Therefore, SCD can be expressed as $\boldsymbol{F}_{\mathrm{SCD}}: \{\boldsymbol{T}_1, \boldsymbol{T}_2\} \rightarrow \{\boldsymbol{P}_{\mathrm{BCD}}, \boldsymbol{S}_1, \boldsymbol{S}_2\}$.

While SCD produces outputs with semantic maps, it primarily relies on visual modality for processing. This lack of direct language input limits its ability to enrich and optimize the understanding and generation processes. Such reliance on a single modality, coupled with challenges in handling complex multioutput scenarios, restricts the broad applicability and effectiveness of SCD in real-world applications. To address this limitation, we reconstructed the SCD category label and extracted the change knowledge. Specifically, give a SCD dataset $\boldsymbol{C}$ containing $\boldsymbol{\tau}$ categories, denoted as $\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3, \ldots, \boldsymbol{c}_t\}$, the construction of change knowledge and category label are illustrated in Fig. 2. The new constructed categories are represented as $\{\boldsymbol{c}_{1\rightarrow 2}, \boldsymbol{c}_{1\rightarrow 3}, \boldsymbol{c}_{1\rightarrow 4}, \ldots, \boldsymbol{c}_{t\rightarrow t-1}\}$, totaling $t \times (t-1)$. Compared with the category labels of SCD, the category labels reconstructed combine the specific change process of two temporal images through combination, making them more relevant to the change detection task. In addition, as the newly constructed category labels

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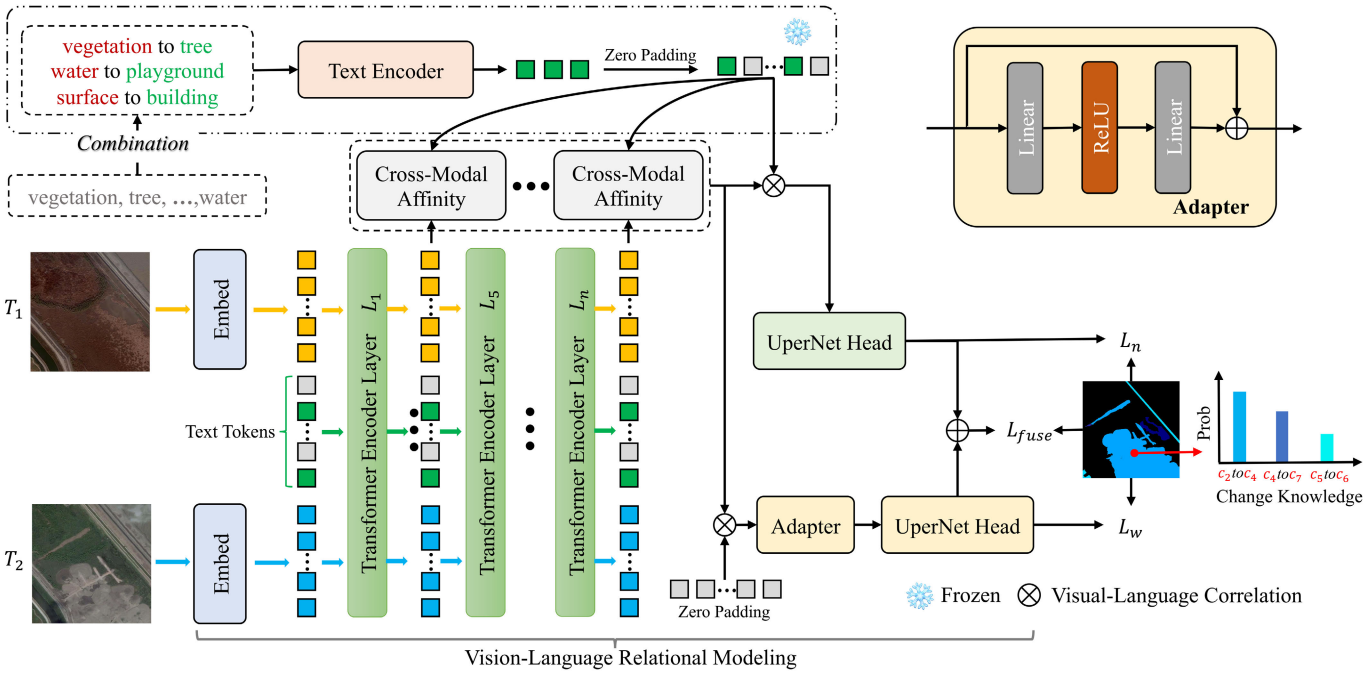


Fig. 3. Overview architecture of our CKCD. We feed two temporal images and task-related change knowledge into the text encoder and patch embedding to generate token embeddings. The subsequent utilization of CMA enhances the model's performance by learning effective complementary features from these two modalities.

encompass both semantic and change information, they can be represented as $\boldsymbol{F}_{\text{CKCD}}: \{\boldsymbol{T}_1, \boldsymbol{T}_2\} \rightarrow \boldsymbol{P}_{\text{CKCD}}$.

Building upon the change knowledge and category label, we introduce the CKCD method, which integrates task-related change knowledge to guide the change detection task. The model's input is expanded to $\{\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{C}_{\text{CKCD}}\}$, where $\boldsymbol{C}_{\text{CKCD}}$ represents the change knowledge. Consequently, CKCD can be expressed as $\boldsymbol{F}_{\text{CKCD}}: \{\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{C}_{\text{CKCD}}\} \rightarrow \boldsymbol{P}_{\text{CKCD}}$.

B. Overall Structure of CKCD

Fig. 3 illustrates the proposed CKCD method, primarily comprised of the visual encoder, text encoder, CMA block, and UperNet Head [55]. Given task-related change knowledge $\boldsymbol{C}_{\text{CKCD}}$ and two temporal images $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$, the text encoder embeds the language information into a specific feature space, generating label embeddings $\boldsymbol{H}'_L \in \mathbb{R}^{N' \times C}$ with $0 < N' \leq t \times (t-1)$. To maintain consistent input lengths across data groups, zero-padded labels are used as placeholders. The resulting padded label embeddings are denoted as $\boldsymbol{H}_L \in \mathbb{R}^{N\text{class} \times C}$, where $N_{\text{class}} = t \times (t-1)$.

For the visual branch, the temporal images $\boldsymbol{T}1$ and $\boldsymbol{T}2$ are first processed through a patch embedding layer. Each image is divided into nonoverlapping patches, flattened, and projected into a $\boldsymbol{D}$-dimensional latent space. This process outputs visual embeddings $\boldsymbol{H}_{T_1} \in \mathbb{R}^{N_{\text{patch}} \times D}$ and $\boldsymbol{H}_{T_2} \in \mathbb{R}^{N_{\text{patch}} \times D}$, where N_{patch} denotes the number of patches per image, and D represents the embedding dimension. These embeddings are concatenated with $\boldsymbol{H}_L$ along the sequence dimension to form the combined feature input $\boldsymbol{H}_{\text{input}} \in \mathbb{R}^{(N_{\text{patch}}+N_{\text{class}}) \times D}$. The combined features are then fed into the visual encoder.

Within the visual encoder, tokens from selected layers (the 4th, 6th, 8th, and 12th layers) are extracted, denoted as

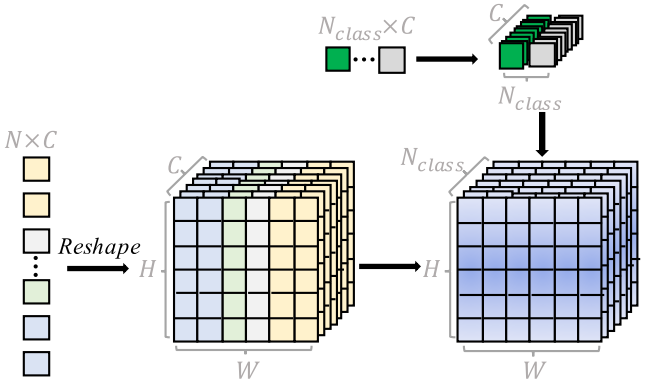


Fig. 4. Detailed design of the proposed V-L Correlation. The features of each pixel in the image are correlated with the corresponding label embedding after padding.

$\boldsymbol{H}^k_{\text{out}} \in \mathbb{R}^{N_k \times D}$ for each layer k, where N_k represents the token count at layer k. These tokens are passed into the CMA blocks alongside $\boldsymbol{H}_L$. The CMA blocks compute cross-modal relationships using attention mechanisms, producing enhanced feature representations $\boldsymbol{H}^k_{\text{CMA}} \in \mathbb{R}^{N_k \times D}$. These features encode both visual and linguistic modalities, highlighting change-related information.

Finally, the enhanced change tokens $\boldsymbol{H}^k_{\text{CMA}}$ are aggregated and correlated with $\boldsymbol{H}_L$ to form a unified representation. This representation is processed by the UperNet Head, which employs a decoder structure to generate the combined semantic change map $\boldsymbol{P}^n_{\text{CKCD}} \in \mathbb{R}^{H \times W \times N_{\text{class}}}$, where H and W are the spatial dimensions of the output map, and N_{class} represents the number of semantic categories, as illustrated in Fig. 4.

In the visual-language correlation (V-L Correlation) process, the change features and the padded language token sequences

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are initially projected to dimensions of $[C \times H \times W]$ and $[C \times N_{\text{class}}]$, respectively. They are then associated through matrix multiplication to generate a tensor of size $[N_{\text{class}} \times H \times W]$. Conversely, for the change detection task that lacks semantic information, zero-padded initialization is used for the learnable parameters in place of semantic information. After being correlated with change features, the resulting tensor undergoes processing through an Adapter module to enhance token representation. Subsequently, it passes through the UperNet Head to generate the change map $\boldsymbol{P}^{w}_{\text{CKCD}}$. Finally, the loss is calculated using the ground truth. Notably, the network parameters associated with the text encoder remain frozen during training.

C. Cross-Modal Affinity

Effectively leveraging information from different modalities is a challenging task [56]. As shown in Fig. 3, the proposed CMA block is used in multiple stages of the visual encoder. The CMA module employs cross-attention mechanisms to correlate visual tokens (e.g., image patches) with textual embeddings (e.g., "water-to-playground"). This process dynamically weights features from both modalities, ensuring context-aware fusion. Formally, given the token sequences of two temporal images $\{\boldsymbol{H}_{T_1}, \boldsymbol{H}_{T_2}\}$ and the padded token sequence of text $\boldsymbol{H}_L$, the designed CMA will learn specific complementary features through these three input streams. The process can be represented as follows:

$$\boldsymbol{F}^i = A^i\left(\boldsymbol{H}^i_{T_1}, \boldsymbol{H}^i_{T_2}, \boldsymbol{H}_L\right), \quad i = 4, 6, 8, 12. \tag{1}$$

Here, A^i represents the ith CMA block, $\boldsymbol{H}^i_{T_1}$ and $\boldsymbol{H}^i_{T_2}$ are the token sequences corresponding to the two temporal images at the ith layer of the visual encoder. The CMA block thoroughly explores and learns complementary information between the visual and language modalities, resulting in more robust feature representations.

The detailed design of the CMA block is shown in Fig. 5. Our CMA block has two branches that are used to attach the intermediate token sequences of the $\boldsymbol{T}_1$ temporal image stream $\boldsymbol{H}^i_{T_1}$ and the $\boldsymbol{T}_2$ temporal image stream $\boldsymbol{H}^i_{T_2}$ separately. Specifically, in the context of the three given input streams, we assign the roles of Query to $\boldsymbol{H}^i_{T_1}$ and $\boldsymbol{H}^i_{T_2}$, denoted as $\boldsymbol{Q}^i_1$ and $\boldsymbol{Q}^i_2$, respectively. Simultaneously, $\boldsymbol{H}_L$ functions as both the Key ($\boldsymbol{K}^i$) and Value ($\boldsymbol{V}^i$). As the parameters of $\boldsymbol{H}_L$ are not optimized (the text encoder remain frozen during training), we employ trainable vectors as prompts for the Key and Value to capture task-specific information at distinct stages. These prompts are represented by $\boldsymbol{P}^i_k \in \mathbb{R}^{N_p \times C}$ and $\boldsymbol{P}^i_v \in \mathbb{R}^{N_p \times C}$, respectively, where N_p indicates the number of prompt tokens. The interactive feature tokens $\boldsymbol{M}^i_1$ and $\boldsymbol{M}^i_2$ for the two branches are computed using the following formula:

$$\boldsymbol{M}^i_1 = \text{CA}\left(\boldsymbol{Q}^i_1, \text{Concat}\left(\boldsymbol{P}^i_k, \boldsymbol{K}^i\right), \text{Concat}\left(\boldsymbol{P}^i_v, \boldsymbol{V}^i\right)\right) \tag{2}$$

$$\boldsymbol{M}^i_2 = \text{CA}\left(\boldsymbol{Q}^i_2, \text{Concat}\left(\boldsymbol{P}^i_k, \boldsymbol{K}^i\right), \text{Concat}\left(\boldsymbol{P}^i_v, \boldsymbol{V}^i\right)\right) \tag{3}$$

in the given formula, $\text{CA}(\cdot, \cdot, \cdot)$ signifies cross attention [57], and $\text{Concat}(\cdot, \cdot)$ represents the concatenation operation along the N dimension. Following this, we project both the interactive features from the two branches and the Query $\boldsymbol{Q}^i_1$ and $\boldsymbol{Q}^i_2$ to the $[C \times H \times W]$ dimension. Enhanced interactive features, denoted as $\boldsymbol{F}^i_1$ and $\boldsymbol{F}^i_2$, are then generated through the gated fusion mechanism [58]

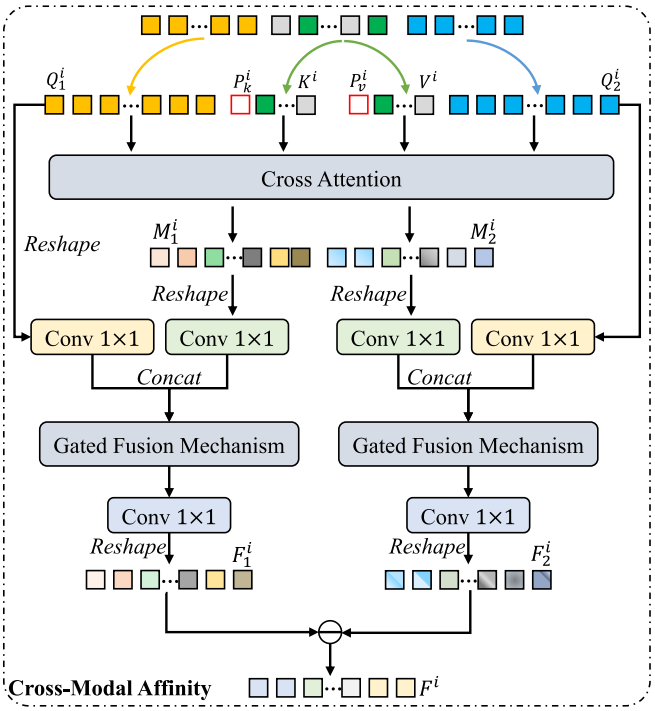


Fig. 5. Detailed design of the proposed CMA. The entire structure is symmetric, handling different temporal images separately. The change features are obtained by computing the difference.

$$\alpha_1 = \text{Sigmoid}\left(\boldsymbol{W}^i_Q \cdot \text{Conv}_1\left(\boldsymbol{Q}^i_1\right) + \boldsymbol{W}^i_M \cdot \text{Conv}_2\left(\boldsymbol{M}^i_1\right)\right) \tag{4}$$

$$\boldsymbol{F}^i_1 = \text{Conv}_3\left((1-\alpha_1) \cdot \boldsymbol{Q}^i_1 + \alpha_1 \cdot \boldsymbol{M}^i_1\right) \tag{5}$$

$$\alpha_2 = \text{Sigmoid}\left(\boldsymbol{W}^i_Q \cdot \text{Conv}_1\left(\boldsymbol{Q}^i_2\right) + \boldsymbol{W}^i_M \cdot \text{Conv}_2\left(\boldsymbol{M}^i_2\right)\right) \tag{6}$$

$$\boldsymbol{F}^i_2 = \text{Conv}_3\left((1-\alpha_2) \cdot \boldsymbol{Q}^i_2 + \alpha_2 \cdot \boldsymbol{M}^i_2\right). \tag{7}$$

In the example, the input channel C is set to 768, and $\boldsymbol{W}^i_Q$ and $\boldsymbol{W}^i_Q$ are learnable parameters. In addition, $\text{Conv}_1(\cdot)$, $\text{Conv}_2(\cdot)$, and $\text{Conv}_3(\cdot)$ represent simple 1×1 convolutions. Ultimately, the change feature $\boldsymbol{F}^i$ is derived by computing the difference between $\boldsymbol{F}^i_1$ and $\boldsymbol{F}^i_2$, the outputs from all four CMA modules are jointly fed into the UperNet head for final prediction.

D. Training and Inference Details

During the training process, CKCD calculates three supervised losses, all based on the weighted cross-entropy (CE) loss function to mitigate the dominance of more frequent categories, thus reducing the model's bias toward majority classes and improving the detection performance for rare change types. Given the ground truth labels $\boldsymbol{Y} \in \mathbb{R}^{H \times W}$ and the predicted semantic change maps $\boldsymbol{P}^{n}_{\text{CKCD}}$, $\boldsymbol{P}^{w}_{\text{CKCD}}$, and $\boldsymbol{P}^{\text{fuse}}_{\text{CKCD}} \in \mathbb{R}^{H \times W \times N_{\text{class}}}$ from different branches, the CE loss for each output is defined as

$$L_n = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{N_{\text{class}}} \boldsymbol{Y}_{i,j,c} \log\left(\boldsymbol{P}^{n}_{\text{CKCD},i,j,c}\right) \tag{8}$$

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$$L_w = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{N_{\text{class}}} \boldsymbol{Y}_{i,j,c} \log\left(\boldsymbol{P}^{w}_{\text{CKCD},i,j,c}\right) \tag{9}$$

$$L_{\text{fuse}} = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{N_{\text{class}}} \boldsymbol{Y}_{i,j,c} \log\left(\boldsymbol{P}^{\text{fuse}}_{\text{CKCD},i,j,c}\right) \tag{10}$$

where $\boldsymbol{Y}_{i,j,c}$ is a one-hot encoded vector representing the ground truth class at pixel position (i, j) and $\boldsymbol{P}_{\text{CKCD},i,j,c}$ denotes the predicted probability for class c at pixel (i, j). The total loss for optimization is the sum of these three components

$$\mathcal{L}_{\text{total}} = L_n + L_w + L_{\text{fuse}}. \tag{11}$$

In the inference process, we introduce two methods to enhance adaptability to practical applications.

1) In instances where task-related change knowledge is present in the test set, each group of temporal images requiring inference possesses known change targets, while the specific change areas remain unknown. In this context, we utilize the fusion result from both branches as the ultimate change prediction map.
2) Conversely, in scenarios where change knowledge is absent in the test set, we exclusively employ the branch with learnable parameters, effectively substituting change knowledge, to generate the final prediction map during the inference stage.

IV. Experiments

Our experiment aims to evaluate and analyze the validity of the proposed CKCD. In this section, we first introduce the experimental settings. Then, we evaluate the results on multiple benchmarks and discuss them. Finally, we conduct exploratory studies of CKCD on the reconstructed SECOND dataset and visualize and analyze the proposed CKCD and the foundation model (RSP-UperNet).

A. Experimental Settings

1) Dataset Description: We evaluate the proposed CKCD method and other state-of-the-art methods on two SCD datasets, SECOND [53] and Hi-UCD [59]. Notably, we have reconstructed category labels for these SCD datasets.

1) *Second:* The SECOND dataset is a well-annotated dataset for SCD. It comprises 4662 pairs of aerial images with a size of 512 × 512, distributed in cities such as Hangzhou, Chengdu, and Shanghai. The dataset focuses on six major land cover categories: nonvegetated ground surface, tree, low vegetation, water, building, and playground. The data is divided into training, validation, and test sets, distributed in a ratio of 6:2:2.
   Then, we reconstruct category labels by combining different semantic categories. The statistics of the newly constructed label categories are shown in Fig. 6. In addition to the unchanged areas, there are 30 newly combined categories, with the largest category accounting for 11% and the nine smallest categories collectively accounting for 10%.
2) *Hi-UCD:* The Hi-UCD dataset contains 359, 386, and 548 pairs of ultrahigh-resolution images captured between 2017 and 2019, with each image measuring 1024 × 1024 pixels and a spatial resolution of 0.1 m. To reduce computational resource overhead, we crop each image into four 512 × 512 images.

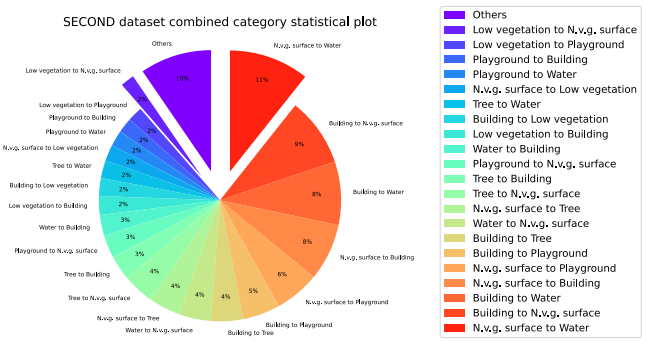


Fig. 6. Pie plot statistics of reconstructed category labels for the SECOND dataset.

In addition, we also validate the proposed CKCD method on BCD datasets, including the LEVIR dataset [25], the WHU dataset [60], and the CDD dataset [61].

1) *LEVIR Dataset:* LEVIR is a large remotely sensed building change detection dataset. Comprising 637 image patch pairs from Google Earth with very high resolution (VHR) at 0.5 m/pixel, each image patch measures 1024 × 1024 pixels. The temporal span of these image pairs ranges from 5 to 14 years. LEVIR covers various buildings, such as villa houses, high-rise apartments, small garages, and large warehouses.
2) *WHU Dataset:* The dataset comprises two aerial images captured in 2012 and 2016, depicting the same 20.5 km^2 area with 12 796 and 16 077 building instances, respectively. The subdataset is geo-corrected to an aerial dataset by manually selecting 30 GCPs at the surface with an accuracy of 1.6 pixels.
3) *CDD Dataset:* The CDD dataset contains three images: synthetic images with no relative displacement of objects, synthetic images with small relative displacement of objects, and real remote sensing images with real seasonal variations (obtained from Google Earth). The remotely sensed images with real seasonal variations contain seven pairs of 4725 × 2700 pixels size from which 16 000 sample pairs of 256 × 256 pixels size (10 000 training set, 3000 test set and 3000 validation set) are collected with a spatial resolution of 3–100 cm.

During training, all BCD datasets are uniformly sized at 256 × 256. We use "No change" and "Change" as the change knowledge in the BCD settings to train the proposed CKCD.

2) Implementation Detail: As our visual encoder, we employ the remote sensing pretrained model RSP [47], while the text encoder is derived from the text feature extractor in CLIP [18]. To maintain fairness in the comparative experiments, the AdamW optimizer [62] is employed during training for all compared methods executed on a single NVIDIA 4090 GPU. We only modify the number of classes in the last layer for all BCD methods, with the

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TABLE I
OVERALL PERFORMANCE ON TWO RECONSTRUCTED SCD DATASETS. WHERE CF IS CONVOLUTIONAL FEATURE, AND TF IS TRANSFORMER. THE TOP THREE HIGHEST SCORES ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	Type	Features	SECOND					Hi-UCD				Param
			Prec.	Rec.	IoU	F1	Kappa	Prec.	Rec.	IoU	F1	
FC-Siam-conc	BCD-based	CF	11.12	22.15	8.50	14.81	30.48	42.26	44.54	30.51	43.37	1.5M
FC-Siam-diff	BCD-based	CF	12.39	21.35	8.12	15.68	30.56	39.98	45.96	29.58	42.76	1.4M
DTCDSCN	BCD-based	CF	4.08	5.10	0.92	4.53	4.18	24.34	13.18	5.41	17.10	41.1M
DSIFN	BCD-based	CF	12.73	16.90	8.31	14.52	45.39	38.91	38.28	27.65	38.29	50.7M
SNUNet	BCD-based	CF	12.30	22.66	9.33	15.94	33.13	43.27	44.06	32.19	43.66	12.0M
USSFCNet	BCD-based	CF	7.29	9.61	4.42	8.29	21.90	27.08	28.53	18.22	27.79	1.5M
HANet	BCD-based	CF	12.61	29.60	9.57	17.69	26.46	48.07	44.85	32.34	46.40	3.1M
SSCFNet	BCD-based	CF	21.01	20.85	13.38	20.93	45.95	50.83	51.82	37.71	50.82	48.2M
P2V-CD	BCD-based	CF	16.89	27.82	11.82	21.02	33.72	41.18	42.44	28.18	41.80	5.4M
ChangeFormer	BCD-based	TF	18.67	24.80	12.99	21.30	39.56	48.49	46.38	34.61	47.41	41.4M
RSP-BIT	BCD-based	TF	10.09	19.64	6.96	13.33	21.88	40.74	33.21	23.31	36.59	97.1M
RSP-UperNet	BCD-based	TF	22.43	23.54	14.95	22.97	48.80	50.80	47.66	36.73	49.18	97.5M
HRSCD	SCD-based	CF	10.79	15.89	6.46	12.86	28.49	37.09	38.47	27.70	37.77	9.3M
Bi-SRNet	SCD-based	CF	22.25	30.90	15.26	24.39	32.37	53.67	50.90	39.85	52.28	23.3M
SSCD-l	SCD-based	CF	22.34	33.48	14.55	25.27	44.73	52.39	55.14	37.78	53.73	23.3M
TED	SCD-based	CF	23.77	35.51	15.85	28.48	37.55	53.53	55.29	40.62	55.90	23.6M
SCanNet	SCD-based	TF	27.00	35.86	19.02	30.87	42.14	55.10	58.44	42.84	56.77	27.9M
CKCD (Ours)	–	TF	32.75	36.77	23.00	34.64	55.60	65.88	58.45	47.56	61.94	99.9M

other structures remaining unchanged. For the compared SCD methods, a semantic segmentation decoder is selected, with any extra decoders removed. We uniformly set the initial learning rate to 4×10^{-5}, with a linear decrease in learning rate, weight decay of 0.01, and 200 epochs of training batches. For training data augmentation, we use random flipping and rotation. The training size for the SECOND and Hi-UCD datasets is 512×512, while the training size for the LEVIR, WHU, and CDD datasets is 256×256.

3) Evaluation Metrics: We adopt multiple widely recognized evaluation metrics to accurately evaluate the algorithm's performance, including precision, recall, intersection over union (IoU), and $F1$-score. These metrics can comprehensively reflect the algorithm's performance in change detection tasks from different perspectives. Since the $F1$-score is the harmonic mean of precision and recall used to evaluate the algorithm's performance comprehensively, we use it as the main evaluation metric for experiments.

B. Main Results

We evaluate the performance of the proposed CKCD on multiple datasets in both reconstructed SCD and BCD datasets. The comparison methods include BCD methods: FC-Siam-conc [7], FC-Siam-diff [7], DTCDSCN [24], DSIFN [26], SNUNet [9], USSFCNet [15], HANet [63], SSCFNet [14], P2V-CD [12], ChangeFormer [33], RSP-BIT [49], and RSP-UperNet [47]; and SCD methods: HRSCD [16], Bi-SRNet [17], SSCD-l [17], TED [54] and SCanNet [54].

1) Comparison of Experiments in Reconstructed SCD Datasets: In the reconstructed SCD datasets, we compare the CKCD (Fuse) method with many representative BCD and SCD methods. Notably, RSP-BIT and RSP-UperNet are methods based on remote sensing pretraining. The evaluation encompasses two datasets, namely, SECOND [53] and Hi-UCD [59], as detailed in Table I. In this challenging new setting, CKCD significantly outperforms other methods on both datasets, achieving $F1$-Score of 34.64% and 61.94%. Compared with the top BCD method, our method improves by 11.67% and 11.12%, respectively. While compared with the best SCD method, our method also improves by 3.77% and 5.17%. It can be inferred that the proposed CKCD effectively utilizes change knowledge to better adapt to more challenging remote sensing images.

2) Comparison of Experiments in BCD: To investigate the performance of CKCD in BCD datasets, we conduct comparisons on the LEVIR [25], WHU [60], and CDD [61] datasets. The results are shown in Fig. 7, indicating that CKCD achieves the highest $F1$-Score of 91.8%, 92.8%, and 97.9% on the three datasets. Compared with RSP-UperNet, CKCD shows improvements of 0.5%, 0.6%, and 0.8%, respectively. However, we observe that the advantage of CKCD is diminished compared to the reconstructed SCD datasets. Our analysis indicates that this reduction is attributed to the BCD setting, which solely contains information about the occurrence of changes, consequently constraining the comprehensive exploitation of CKCD's cross-modal interaction capabilities.

3) Category Radargrams: Fig. 8 shows the evaluation results based on the reconstructed SECOND datasets, indicating that our CKCD method outperforms other BCD and SCD methods in almost all change categories. In particular,

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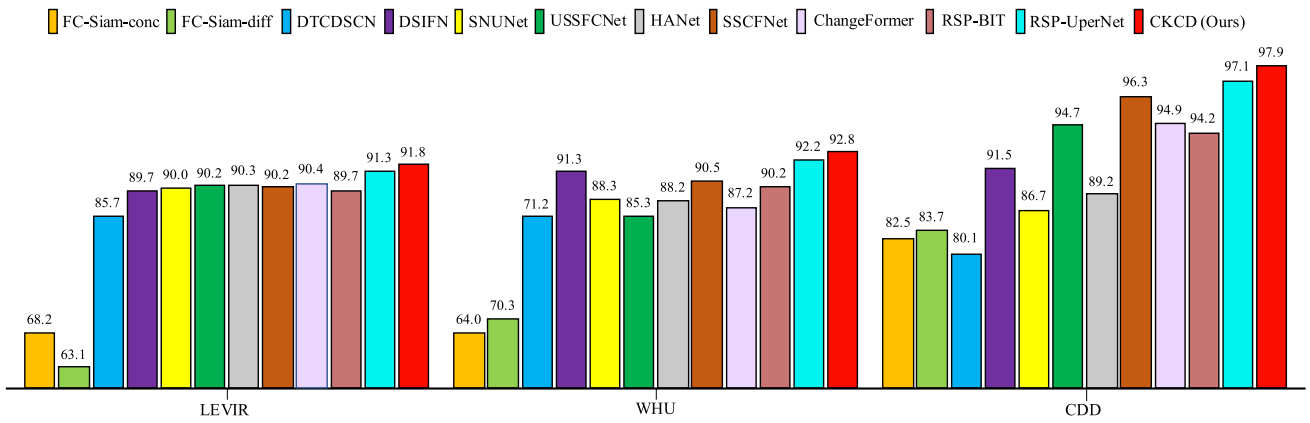


Fig. 7. State-of-the-art comparison on LEVIR, WHU, and CDD datasets.

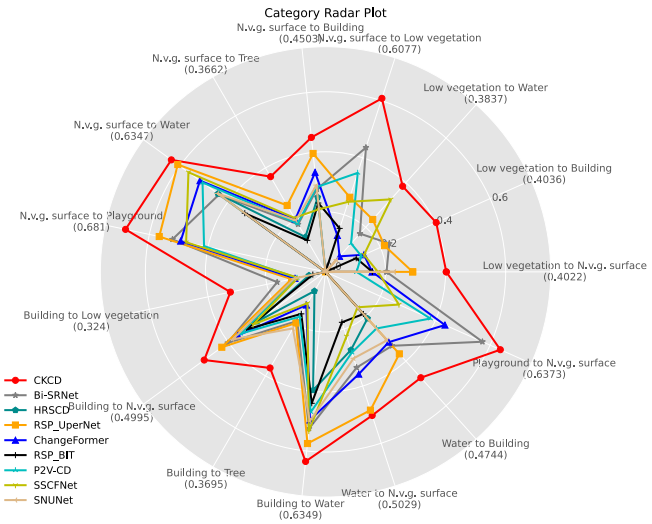


Fig. 8. $F1$-scores for different categories on the reconstructed SECOND dataset.

it has a significant advantage in the combined categories with large temporal image changes, such as "Nonvegetated ground surface to Low vegetation," "Nonvegetated ground surface to Playground," "Building to Low vegetation," and "Low vegetation to Building," demonstrating the superior discriminative ability of CKCD.

4) Qualitative Comparison: Fig. 9 presents a qualitative comparison of four representative datasets from the reconstructed SECOND datasets. Compared with other methods, the proposed CKCD performs better, with detection results exhibiting relatively clear and accurate boundaries and a significantly lower false detection rate than other methods.

C. Exploration Studies

We analyze the main properties of the CKCD method on the reconstructed SCD datasets. Unless otherwise noted, all experiments in the following study are conducted on the SECOND dataset. The default settings are marked in gray.

1) Impact of Different Components: To evaluate the contribution of each component in the proposed CKCD, we conduct an ablation study using four variants of CKCD and the foundation model (RSP-UperNet), as shown in Table II. The foundation model's baseline performance, with an IoU of 14.95% and an $F1$-score of 22.97%, reflects limited capacity in capturing both visual and semantic change features. Introducing the CMA block improves IoU and $F1$ by 3.11% and 4.95%, respectively, demonstrating its effectiveness in enhancing the model's ability to integrate task-related change knowledge, reducing modality gaps, and improving the clarity of change region identification, particularly in complex scenes where visual clues alone may be ambiguous. Adding the V-L Correlation module further enhances IoU and $F1$ to 20.45% and 30.56% by aligning visual and language features more effectively, strengthening semantic understanding, and ensuring the language descriptions better reflect the corresponding visual patterns for clearer change region distinction. In addition, the Adapter module refines feature representation by introducing lightweight learnable layers between the visual encoder and subsequent components, enhancing cross-modal feature alignment and further improving detection reliability. Since the text encoder parameters remain frozen during training, a key-value prompt mechanism is employed to dynamically adjust language features, leading to the best overall performance, with IoU and $F1$ increasing by 8.05% and 11.67%, respectively, compared to the foundation model. These results collectively demonstrate how the progressive integration of CMA, V-L Correlation, adapter, and key-value prompt significantly enhances both spatial localization and semantic understanding, contributing to a more comprehensive and accurate change detection process.

2) Study of Scores in Each Category: The confusion matrix of categories is plotted to explore the scores of each combined category further. In the confusion matrix, the deeper the color, the higher the corresponding category's score. As shown in Fig. 10, we observe that the proposed CKCD generally outperforms RSP-UperNet. Among the categories, "Playground to N.v.g. surface" has the highest score of 68.1%. Please note that since there are no changes in all the same categories, the diagonal scores are all 0.

3) Exploration of Different Outputs: We analyze the different outputs of the proposed CKCD method. CKCD (weak)

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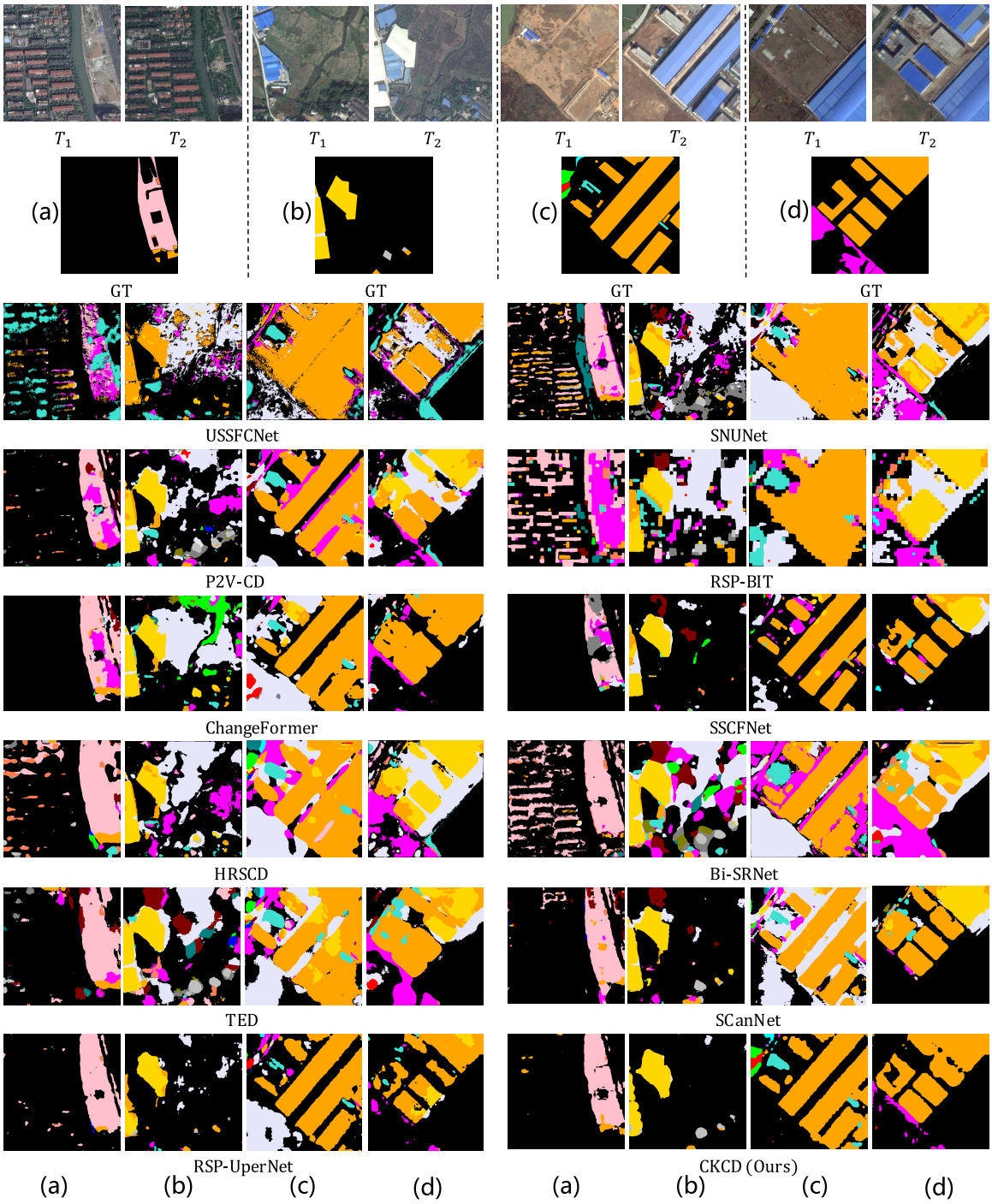


Fig. 9. Qualitative comparison results of the proposed algorithm and nine state-of-the-art methods on four sets of test data. Our CKCD performs well with the help of change knowledge. (a)–(d) Represents four different representative samples in the dataset.

TABLE II

QUANTIFYING THE CONTRIBUTION OF DIFFERENT COMPONENTS IN CKCD TO THE RECONSTRUCTED SECOND DATASETS. (a)–(c) REFER TO DIFFERENT ABLATION STUDY VARIANTS OF THE PROPOSED CKCD, REPRESENTING DISTINCT EXPERIMENTAL SETUPS

Method	CMA	V-L Correlation	Adapter	Key-Value Prompt	IoU (↑)	F1 (↑)
Foundation model (RSP-UperNet)					14.95	22.97
(a)	✓				$18.06_{+3.11}$	$27.92_{+4.95}$
(b)	✓	✓			$20.45_{+5.50}$	$30.56_{+7.59}$
(c)	✓	✓	✓		$21.26_{+6.31}$	$32.12_{+9.15}$
CKCD (Ours)	✓	✓	✓	✓	$\mathbf{23.00}_{+8.05}$	$\mathbf{34.64}_{+11.67}$

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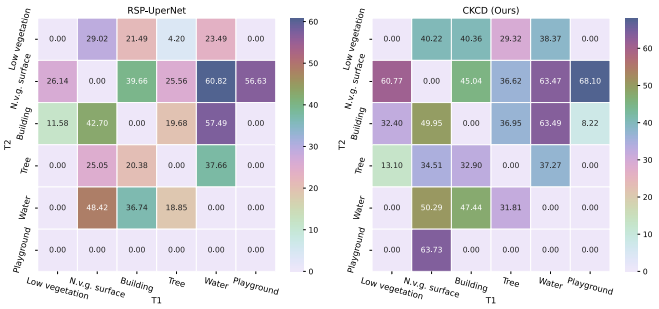


Fig. 10. $F1$-scores for each category on the reconstructed SECOND datasets. The left figure shows the foundation model RSP-UperNet, and the right figure shows the proposed CKCD.

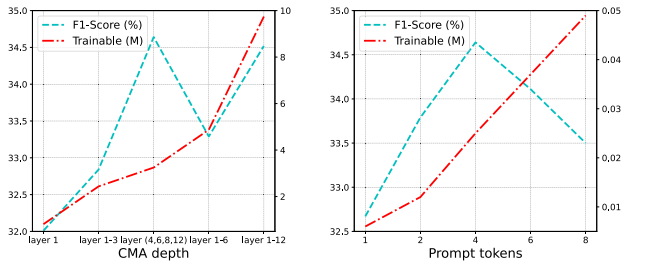


Fig. 11. (Left) Ablation of CMA depth and (right) key-value prompt length in CKCD.

TABLE III

EXPLORATION OF DIFFERENT OUTPUTS OF THE PROPOSED CKCD ON THE RECONSTRUCTED SECOND DATASETS

Method	IoU (↑)	F1 (↑)
CKCD (Weak)	21.46	32.63
CKCD (Normal)	22.82$_{+1.36}$	33.93$_{+1.30}$
CKCD (Fuse)	**23.00**$_{+1.54}$	**34.64**$_{+2.01}$

TABLE IV

ABLATION OF THE PROPOSED CKCD WITH DIFFERENT PRETRAINING

Method	IoU (↑)	F1 (↑)
CKCD (IMP)	22.20	33.51
CKCD (RSP)	**23.00**$_{+0.80}$	**34.64**$_{+1.13}$

TABLE V

STUDY OF THE PROPOSED CKCD WITH DIFFERENT KEY-VALUE PROMPT WAYS

Method	IoU (↑)	F1 (↑)
Key-Value Prompt (Replace)	22.56	33.32
Key-Value Prompt (Concat)	**23.00**$_{+0.44}$	**34.64**$_{+1.32}$

(first row) represents the output generated by replacing the language information branch with learnable parameters. CKCD (normal) (second row) represents the normal output. CKCD (fuse) (third row) represents the fusion result of the first two outputs. As shown in Table III, we observe that the fusion output result of CKCD (fuse) achieves the best score. Although not using task-related language information, CKCD (weak) only has IoU and $F1$-scores that are 1.54% and 2.01% lower than CKCD (fuse), indicating that using learnable parameters to replace language information is effective.

4) Ablation of Different Pretraining: We further conduct ablation studies on different pretraining options. IMP represents the visual encoder initialized with MAE [64] pretraining parameters, and RSP represents the visual encoder initialized with remote sensing pretraining parameters. As shown in Table IV, RSP is more suitable for the current change detection task than IMP.

5) Study of Different Key-Value Prompt Ways: Table V lists the impact of the proposed CKCD on the reconstructed second dataset under different key-value prompt ways. "Replace" represents using learnable parameters to replace some of the key and value parameters while keeping the number of tokens unchanged. "Concat" represents concatenating learnable parameters while keeping the key and value parameters fixed, increasing the number of tokens. After comparison, it can be found that the "Concat" achieves higher performance.

6) Number Analysis of CMA: The CKCD method incorporates CMA blocks into different semantic layers to establish cross-modal feature affinity. In this study, we systematically explored the impact of adjusting the number of CMA blocks, employing two insertion strategies. The first strategy entails sequentially inserting CMA blocks from shallow to deep layers. In contrast, the second involves inserting CMA blocks at the outputs of layers 4, 6, 8, and 12, aligning with the feature pyramid structure formed by UperNet. As shown in Fig. 11(left), for the first strategy, the model performance and the trainable parameters increase stably with the number of inserted CMA blocks. In contrast, with the second strategy, not only does the performance surpass the optimal performance achieved with the first strategy, but the number of trainable parameters also remains within an acceptable range.

7) Prompt Length: Fig. 11(right) illustrates the impact of Key-Value Prompt length on the performance of CKCD. The optimal performance for CKCD is achieved when the prompt length is 4. Our analysis indicates that when the prompt length is too low, the learning ability of the prompt parameter is limited. Conversely, when the prompt length is too long, the prompt parameter may learn excessive redundancy and invalid knowledge, affecting model performance.

D. Visualization and Analysis

To explore the working principle of language information in CKCD, we conduct ablation experiments on several representative data from the SECOND dataset. The visual results are shown in Fig. 12. We compare the feature responses of the visual encoder layers 4, 6, 8, and 12 of CKCD and the foundation model. CKCD can better capture object edges and details in shallow features. CKCD's feature maps show strong global information integration ability in deep features, enabling accurate change detection across the entire scene. In contrast, the feature maps of the foundation model are relatively chaotic and difficult to distinguish different objects. Therefore, through the affinity of visual and language features in the CMA block, it is clear that CKCD excels in detail capture, semantic information integration, and contextual information integration, and exhibits strong global information integration ability at the

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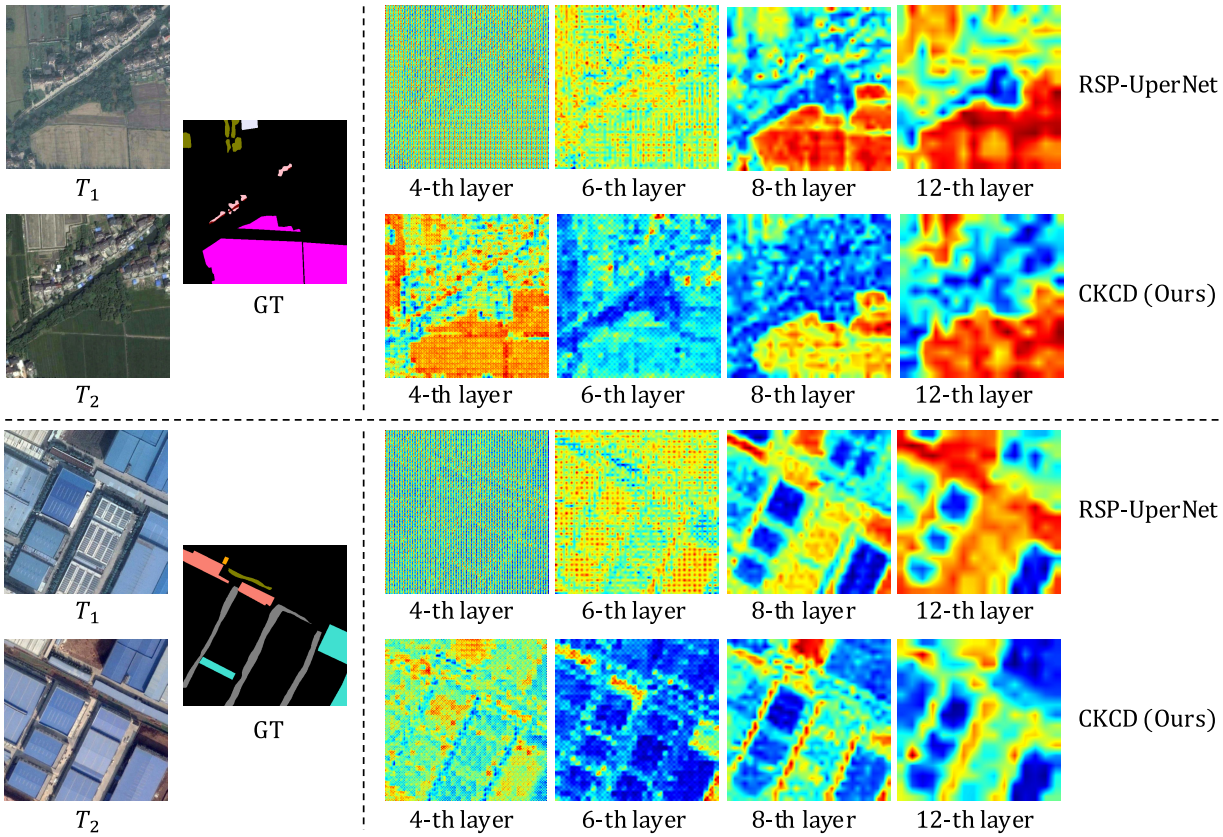


Fig. 12. Visualization results of feature response maps of the proposed CKCD and the foundation model (RSP-UperNet) on two test datasets. Columns 3–6 show the feature visualization results of the visual encoder's 4th, 6th, 8th, and 12th layers. Best viewed in color.

highest level. These advantages make CKCD perform better in change detection tasks.

V. Conclusion

This work introduces the CKCD method, which effectively integrates change knowledge to enhance both the coherence and efficacy of remote sensing change detection tasks. By leveraging this change knowledge, our approach improves the transparency and interpretability of the decision-making process, making the results more comprehensible. In addition, we proposed the CMA module to address the discrepancies between visual and textual modalities, thereby enhancing information complementarity and ensuring consistency throughout the fusion process. Our experimental results demonstrate that CKCD performs better than state-of-the-art methods on two SCD datasets incorporating change knowledge and multiple BCD datasets.

Future work will focus on further enhancing multimodal information fusion and exploring the application of CKCD in other remote sensing tasks to improve generalizability and applicability across different domains. And we plan to explore more lightweight architectures, such as using dynamic networks or efficient attention mechanisms, to reduce the model complexity while maintaining performance. In addition, strategies like knowledge distillation or pruning could be investigated to further optimize the model for resource-constrained environments.

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Jiahao Wang received the B.S. degree in computer science and technology from the North University of China, Taiyuan, China, in 2021. He is currently pursuing the Ph.D. degree in computer science and technology with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His main research interests include remote sensing image processing and visual tracking.



Fang Liu (Senior Member, IEEE) received the B.S. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 1984, and the M.S. degree in computer science and technology from Xidian University, Xi'an, in 1995.

She is currently a Professor with the School of Computer Science, Xidian University. Her research interests include signal and image processing, synthetic aperture radar image processing, multiscale geometry analysis, learning theory and algorithms, optimization problems, and data mining.

Prof. Liu won the Second Prize of the National Natural Science Award in 2013.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a Foreign Member of the Academia Europaea and Russian Academy of Natural Sciences; a fellow of IET, CAAI, CIE, CCF, and CAA; a Councilor of Chinese Institute of Electronics; a Committee Member of Chinese Committee of Neural Networks; and an Expert of the Academic Degrees Committee of the State Council; the Chairperson of the Awards and Recognition Committee; and the Vice Board Chairperson of Chinese Association of Artificial Intelligence.



Hao Wang received the B.S. degree in computer science and technology from the North University of China, Taiyuan, China, in 2021. He is currently pursuing the Ph.D. degree in computer science and technology with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His main research interests include remote sensing image processing and computer vision.



Shuo Li (Member, IEEE) received the B.S. degree in software engineering from Wuhan University, Wuhan, China, in 2016, and the Ph.D. degree in computer science and technology from Xidian University, Xi'an, China, in 2023.

He is currently a Post-Doctoral Researcher with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University. His research interests include computer vision, pattern recognition, and image interpretation.



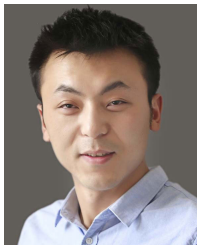
Lingling Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2017, respectively.

From 2013 to 2014, she was an Exchange Ph.D. Student with the Intelligent Systems Group, Department of Computer Science and Artificial Intelligence, University of the Basque Country (UPV/EHU), Leioa, Spain. She is currently a Post-Doctoral Researcher with the School of Artificial Intelligence, Xidian University. Her current research interests include quantum evolutionary optimization, machine learning, and deep learning.



Puhua Chen (Senior Member, IEEE) received the B.S. degree in environmental engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2009, and the Ph.D. degree in circuit and system from Xidian University, Xi'an, China, in 2016.

She is currently an Associate Professor with the School of Artificial Intelligence, Xidian University. Her current research interests include machine learning, pattern recognition, and remote sensing image interpretation.



Xu Liu (Senior Member, IEEE) received the B.S. degree in mathematics and applied mathematics from the North University of China, Taiyuan, China, in 2013, and the Ph.D. degree from Xidian University, Xi'an, China, in 2019.

He is currently an Associate Professor at Huashan Elite, Xi'an, and a Post-Doctoral Researcher with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University. His research interests include machine learning and image processing.

Dr. Liu was the Chair of the IEEE Xidian University Student Branch from 2015 to 2019.



Wenping Ma (Senior Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently an Associate Professor with the School of Artificial Intelligence, Xidian University. Her research interests include natural computing and intelligent image processing.

Dr. Ma is a member of CIE.

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